

Git Hands-on Lab - Exercise 4 (Merge Conflict Resolution)

Exercise: Merge Conflict Resolution

Aim

To understand how merge conflicts occur in Git and how to resolve them by merging changes from different branches.

Procedure and Commands

1. Navigate to project folder
`cd /e/DEEPSKILLING/"MODULE 11 GIT"/GitConflictDemo`
2. Initialize repository
`git init`
3. Rename default branch
`git branch -M main`
4. Create README
`touch README.md`
5. Stage README
`git add README.md`
6. Initial commit
`git commit -m "Initial commit"`
7. Create branch
`git branch GitWork`
8. Switch to branch
`git checkout GitWork`
9. Create hello.xml
`echo "<message>Hello from GitWork Branch</message>" > hello.xml`
10. Stage hello.xml
`git add hello.xml`
11. Commit branch changes
`git commit -m "Added hello.xml in GitWork"`
12. Switch to main
`git checkout main`
13. Create conflicting hello.xml
`echo "<message>Hello from Main Branch</message>" > hello.xml`

14.Stage file

```
git add hello.xml
```

15.Commit main changes

```
git commit -m "Added hello.xml in main"
```

16.View history

```
git log --oneline --graph --decorate --all
```

17.Compare branches

```
git diff main GitWork
```

18.Merge branch

```
git merge GitWork
```

19.Resolve conflict

Edit hello.xml manually and remove conflict markers.

20.Stage resolved file

```
git add hello.xml
```

21.Commit resolved merge

```
git commit -m "Resolved merge conflict"
```

22.Create .gitignore

```
touch .gitignore
```

23.Add *.bak entry

Open .gitignore and add: *.bak

24.Commit .gitignore

```
git add .gitignore && git commit -m "Added .gitignore"
```

25.Delete merged branch

```
git branch -d GitWork
```

26.Push to GitHub

```
git push -u origin main
```

Expected Result

The merge conflict is resolved successfully, the changes are committed, the .gitignore file is added, the temporary branch is deleted, and the repository is pushed to GitHub.

- git branch GitWork

```
vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitConflictDemo (main)
$ git branch GitWork

vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitConflictDemo (main)
$ git branch
  GitWork
* main
```

- git status

```
vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitConflictDemo (GitWork)
$ git status
On branch GitWork
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        hello.xml

nothing added to commit but untracked files present (use "git add" to track)
```

- git log --oneline --graph --decorate --all

```
vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitConflictDemo (main)
$ git log --oneline --graph --decorate --all
* 2cb8a03 (HEAD -> main) Added hello.xml in main
| * 2464179 (GitWork) Added hello.xml in GitWork
|/
* 5b57e65 Intial commit
```

- git diff main GitWork

```
vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitConflictDemo (main)
$ git diff main GitWork
diff --git a/hello.xml b/hello.xml
index 064b23b..234c289 100644
--- a/hello.xml
+++ b/hello.xml
@@ -1,1 @@
- <message>Hello from Main Branch</message>
+ <message>Hello from Gitwork Branch</message>
```

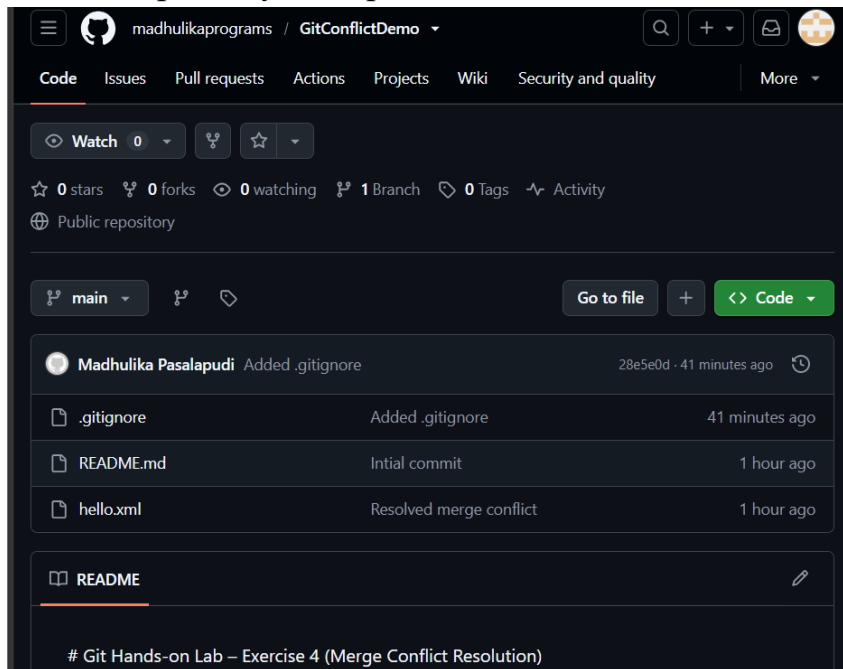
- Merge conflict in hello.xml
- Resolved hello.xml

```
vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitConflictDemo (main|MERGING)
$ git commit -m "Resolved merge conflict"
[main 9827c9d] Resolved merge conflict
```

- git commit after resolving conflict
- git branch -d GitWork

```
vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitConflictDemo (main)
$ git branch -d GitWork
Deleted branch GitWork (was 2464179).
```

- GitHub repository after push



Conclusion

The merge conflict was successfully resolved, the repository history was updated, the temporary branch was deleted, and the project was successfully pushed to GitHub.